ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PRACTICE EXAM 2

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Paper 2 (solution development). Section A: objective questions (30) · Section B: hardware and software, programming tools (compiling and building), C programs with printf and scanf, data types and arithmetic, decisions and loops, and Scratch for the Raspberry Pi (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write all program code neatly. Small spelling mistakes in code will not be penalised if the logic is correct.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The part of the computer that carries out program instructions is the …

A hard drive. B central processing unit (CPU).
C monitor. D power supply.

1.1.2 Which of the following is system software?

A A word processor B A game
C An operating system D A spreadsheet

1.1.3 C source files normally have the extension …

A .c B .py C .exe D .txt

1.1.4 Changing C source code into machine code is called …

A debugging. B compiling.
C commenting. D formatting.

1.1.5 The C function used to display output on the screen is …

A scanf() B input() C cout D printf()

1.1.6 Which data type stores a whole number in C?

A int B float C char D double

1.1.7 In C, the result of 17 % 5 is …

A 3 B 3,4 C 2 D 12

1.1.8 The format specifier used with printf() to show a float is …

A %d B %c C %f D %s

1.1.9 A loop that always runs at least once is a …

A do…while loop. B for loop.
C while loop. D if statement.

1.1.10 In Scratch on a Raspberry Pi, GPIO pins are controlled with blocks from the …

A Sound category only.
B Looks category only.
C Pen extension.
D Raspberry Pi GPIO extension.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 RAM keeps its contents when the computer is switched off.

1.2.2 An IDE such as Geany lets you write, compile and run programs in one place.

1.2.3 A syntax error stops a C program from compiling.

1.2.4 Every C statement normally ends with a semicolon.

1.2.5 In C, = compares two values and == assigns a value.

1.2.6 A variable must be declared before it is used in C.

1.2.7 scanf("%d", &age); reads a whole number into the variable age.

1.2.8 Comments are ignored by the compiler.

1.2.9 A while loop checks its condition before every repetition.

1.2.10 The "forever" block in Scratch repeats the blocks inside it only once.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Physical parts of a computer	A	variable
1.3.2	Step-by-step set of instructions to solve a problem	B	logic error
1.3.3	A named storage place for a value	C	trace table
1.3.4	Finding and fixing mistakes in a program	D	#include
1.3.5	File produced by the build process that can be run	E	constant
1.3.6	Error where the program runs but gives a wrong answer	F	debugging
1.3.7	Line that adds a library such as stdio.h	G	executable file
1.3.8	Table used to follow variable values step by step	H	hardware
1.3.9	Value that does not change while the program runs	I	algorithm
1.3.10	Function where every C program starts	J	main()

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: HARDWARE, SOFTWARE AND TOOLS

2.1 Name TWO input devices, TWO output devices and TWO storage devices. (6)

2.2 Differentiate between system software and application software, and give ONE example of each. (4)

2.3 Explain the difference between compiling and building a C program in Geany, and name the type of file each step produces. (4)

2.4 Name and explain the THREE types of programming errors. (6)

[20]

QUESTION 3: A C PROGRAM

3.1 Study the C program below and answer the questions.

```
#include <stdio.h>

int main(void) {
    float voltage, current, resistance;

    printf("Enter the voltage (V): ");
    scanf("%f", &voltage);
    printf("Enter the current (A): ");
    scanf("%f", &current);

    resistance = voltage / current;
    printf("Resistance = %.2f ohm\n", resistance);
    return 0;
}
```

3.1.1 What is the purpose of #include <stdio.h>? (1)

3.1.2 Why are the variables declared as float and not int? (1)

3.1.3 What does the & in front of voltage mean in scanf()? (1)

3.1.4 What will the program display if the user enters 12 and 0,5 (typed as 0.5)? (2)

3.1.5 What problem will happen if the user enters 0 for the current? Write an if-statement that prevents it. (4)

3.2 Write a complete C program that asks for the number of hours a robot works per day and the cost of electricity per hour, then calculates and displays the cost for a 5-day week. (8)

[17]

QUESTION 4: DECISIONS AND LOOPS

4.1 Study the code below.

```
int i, total = 0;
for (i = 1; i <= 5; i++) {
    if (i % 2 == 0) {
        total = total + i;
    }
}
printf("%d", total);
```

4.1.1 Draw a trace table showing the values of i and total. (5)

4.1.2 What is displayed? (1)

4.1.3 Rewrite the loop as a while loop. (4)

4.2 Write C code that reads a test mark and displays "Distinction" for 80 and above, "Pass" for 50 to 79, and "Fail" below 50. If the mark is less than 0 or more than 100 it must display "Invalid mark". (7)

[17]

QUESTION 5: SCRATCH AND THE RASPBERRY PI

5.1 An LED is connected to GPIO 18 and a push button to GPIO 4 of a Raspberry Pi. Describe, block by block, a Scratch script that switches the LED on while the button is pressed and off when it is released. (6)

5.2 Explain why block-based languages such as Scratch are useful for beginners. Give TWO reasons. (2)

5.3 Draw a flowchart for a traffic light that shows red for 5 seconds, green for 4 seconds and amber for 2 seconds, and repeats forever. (8)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER